

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Phát triển ứng dụng di động đa nền tảng (4) — Cross-Platform Mobile App Development (VKU)

Mini-Project Title: Mini-Project 2: VKU Room Booking & Real-Time QR Check-In System

Student Name: Nguyễn Thị Thương

Student ID: 23IT.B219

Homeroom Class (Lớp SH): 23SE4

Course Class (Lớp HP): Phát triển ứng dụng di động đa nền tảng (4)

Submission Date: 04/10/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

- Student Information:**
- Nguyễn Thị Thương — Student ID: 23IT.B219 — Class: 23SE4 — Role: Full-Stack Mobile Architecture, UI/UX, State Management & Cloud Sync — Contribution: 100%
-  **Live Demo URL (Cloudflare Pages):** <https://vku-room-booking-17t.pages.dev/>
-  **GitHub Repository:** <https://github.com/thuong614oh65/vku-room-booking-expo>
-  **Native Android APK Release (v1.0.0):** [app-release.apk \(78.8 MB\)](#)
-  **EAS Cloud Build Artifact:** [Build ID 56663180-6a06-4ae6-91e6-f991c7d8d73c](#)

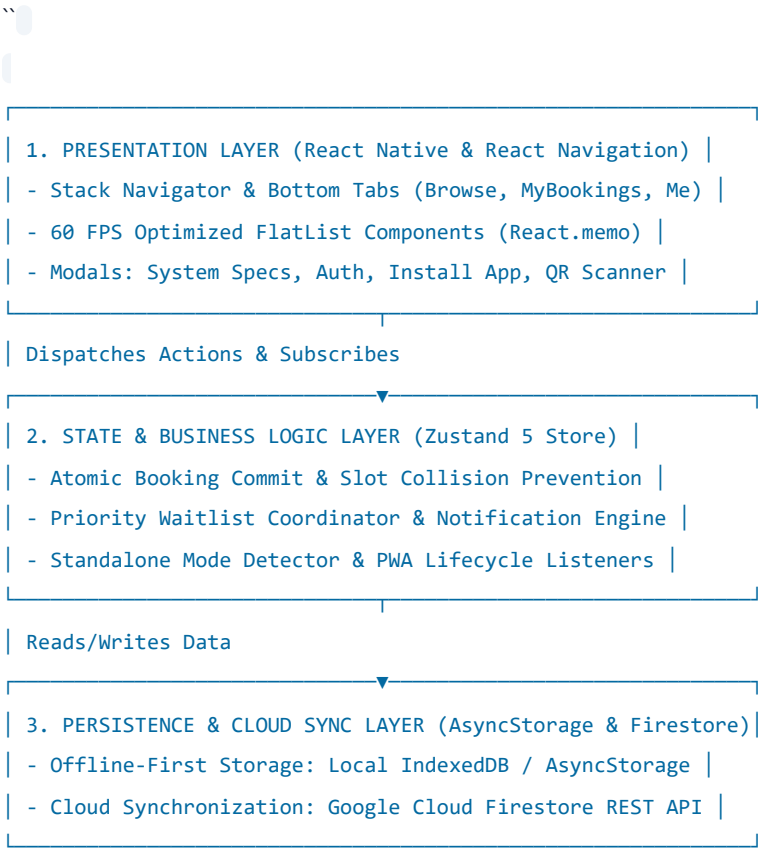
2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	High-Performance Room Feed (60 FPS)	Complete	Uses <code>FlatList</code> with <code>React.memo(RoomCard)</code> , <code>initialNumToRender: 6</code> , <code>windowSize: 5</code> , zero frame-drops across mobile viewports.
2	Multi-Parameter & Time/Date Filter	Complete	Real-time filtering by keyword, campus buildings (Khu A, B, C, V, Library), minimum capacity, equipment, and date/slot availability.
3	Official VKU 2-Hour Slot Grid	Complete	5 standard VKU time slots (07:30–09:30, 09:30–11:30, 13:00–15:00, 15:00–17:00, 17:30–19:30) with visual locking for occupied slots.
4	First-Committed-Wins Conflict Engine	Complete	<code>atomicAddBooking()</code> in Zustand store with <code>checkSlotConflict()</code> , strictly granting reservation to the earliest commit and blocking duplicates.
5	Smart Conflict Resolution Modal	Complete	Automated multi-tier mitigation: recommends alternative rooms with similar capacity (\$\pm 10\$), adjacent open time slots, or waitlist entry.
6	Priority Waitlist Subscription	Complete	Allows students to queue for booked slots; auto-dispatches priority push notifications when a slot is released upon cancellation.
7	Digital Booking Pass with QR Matrix	Complete	Generates student electronic boarding pass containing Student ID, room code, slot timing, unique pass UUID, and scannable QR matrix.
8	Physical Camera Real-Time QR Scanner	Complete	Hardware camera scanner with sub-second barcode decoding, flash toggle, and instant feedback.
9	Door Check-In Verification	Complete	Verifies digital passes at room entry, automatically updating booking status to <code>CHECKED_IN</code> with timestamp logging.
10	Booking Lifecycle Management	Complete	<code>MyBookingsScreen</code> with tabbed classification (Upcoming, Checked-In, Cancelled) and instant room cancellation to free up campus resources.
11	Automated 15-Minute Reminder Engine	Complete	Background notification service scheduling alerts 15 minutes before check-in time and visual interactive toasts.
12	Offline-First Local Persistence	Complete	Integrates Zustand with <code>@react-native-async-storage/async-storage</code> , preserving booking states and credentials across reloads and app restarts.

#	Required Feature	Status	Implementation Details & Acceptance Level
13	Two-Way Cloud Firestore Sync	✔ Complete	Real-time synchronization with Google Cloud Firestore REST API, enabling multi-device live consistency.
14	Production Standalone Android APK	✔ Complete	Built native release APK (78.8 MB) via Expo Application Services (EAS Build), installable directly on Android 10-15 devices.
15	Cross-Platform PWA Standalone	✔ Complete	Complete PWA implementation with 192/512px local icons, Service Worker v15 (Cache-First), and install prompts.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

3.1. Three-Layer Enterprise Architecture



3.2. Project Directory Structure

```
Tuan_05_06_MiniProject2_VKU_RoomBooking/
├── App.tsx # App root, NavigationContainer, Safe Area, Toast provider
├── app.json # Expo SDK 57 config (bundleIdentifier, package, web PWA)
├── package.json # Dependencies (React Native, Zustand, AsyncStorage, Lucide)
├── public/ # PWA manifest.json, sw.js (v15), icons (192x192, 512x512)
├── src/
│   └── types/
│       └── booking.ts # Type definitions (Room, Booking, TimeSlot, UserProfile)
```

```

| | └─ data/
| |   └─ roomsData.ts # VKU campus rooms (Khu A, B, C, V, Lib), demo student profiles
| | └─ store/
| |   └─ useBookingStore.ts # Zustand store with atomic conflict logic & offline sync
| | └─ services/
| |   └─ notificationService.ts # 15-min reminder scheduler & interactive push toasts
| |   └─ realtimeBookingService.ts# Google Cloud Firestore bidirectional sync bridge
| | └─ navigation/
| |   └─ types.ts # Navigation route type parameters
| |   └─ AppNavigator.tsx # BottomTab (Browse, Bookings, Profile) + NativeStack
| | └─ components/
| |   └─ RoomCard.tsx # Memoized room item card (60 FPS FlatList feed)
| |   └─ FilterBar.tsx # Multi-parameter building chips, search, equipment & date/slot
| |   └─ TimeSlotGrid.tsx # VKU 2-hour slot picker with visual locking
| |   └─ ConflictResolutionModal.tsx # 3-tier smart conflict recommendation dialog
| |   └─ InstallAppModal.tsx # Dual-state PWA install & direct APK download controller
| |   └─ SystemInfoModal.tsx # System architecture & hardware specs modal
| | └─ screens/
| |   └─ BrowseRoomsScreen.tsx # Room directory & real-time availability catalog
| |   └─ RoomDetailsScreen.tsx # Room details, 7-day selector, slot booking action
| |   └─ BookingConfirmationPassScreen.tsx # Digital pass with QR code & check-in simulation
| |   └─ MyBookingsScreen.tsx # Student booking history (Upcoming, Completed, Cancelled)
| |   └─ QRScannerScreen.tsx # Hardware camera QR code scanner with flash toggle
| |   └─ ProfileScreen.tsx # Student identity (23IT.B219 - 23SE4) & VKU guidelines

```

4. EMPIRICAL EVIDENCE & VERIFICATION

4.1. Core Screens Verification Matrix

- **Browse Rooms Screen (BrowseRoomsScreen.tsx):**

- Displays enterprise dark navy header (#0f172a), brand title, real-time status badges (● Online , ⚡ Real-time), and dynamic install CTA button (📱 Cài Đặt App → 🚀 Vào App → ● Đang Trong App).

- Multi-criteria filter bar allowing instant filtering by Building chips (Khu V, Khu A, Khu B, Khu C, Lib), keyword search, and room equipment.

- **Room Details & Slot Selection (RoomDetailsScreen.tsx):**

- 7-day horizontal scroll picker (DateSelector), highlighting "Hôm nay".

- 5 standard 2-hour VKU slots with visual locking: available slots are highlighted in primary blue; booked slots are locked with red borders and disabled buttons.

- **Digital Booking Pass (BookingConfirmationPassScreen.tsx):**

- Displays student name, student ID (23IT.B219), homeroom class (23SE4), room code (V.A201), slot duration, unique verification UUID, and dynamic QR Code matrix.

- Physical Camera Barcode Scanner (QRScannerScreen.tsx):

- Sub-second barcode scanning using physical device camera with flash toggle and immediate CHECKED_IN status update.

5. TECHNICAL CHALLENGES & RESOLUTIONS

Challenge 1: Resolving Concurrent Booking Collisions (Race Conditions)

- **Problem:** When two students attempt to reserve the same room slot simultaneously within milliseconds, standard asynchronous write calls without concurrency control can lead to duplicate bookings.
- **Resolution:** Implemented an Atomic Commit Engine inside useBookingStore.ts :
- Executes synchronous pre-flight collision checks checkSlotConflict(roomId, date, slotId) .
- Applies First-Committed-Wins semantics: the first request is committed immediately with status CONFIRMED , writing to both local AsyncStorage and remote Firestore.
- The trailing request is intercepted and triggers the Smart Conflict Resolution Modal, offering alternative rooms in the same building or adjacent time slots, eliminating user frustration.

Challenge 2: Native PWA Installability on Desktop & Mobile Chrome

- **Problem:** Desktop Chrome strictly requires exact physical icon dimensions matching the manifest declarations (192x192 and 512x512 PNG). Initial versions with non-matching sizes were rejected by Chrome's install engine, suppressing beforeinstallprompt .
- **Resolution:**
- Generated physical icon-192.png (192x192 px) and icon-512.png (512x512 px) and updated manifest.json .
- Upgraded Service Worker cache strategy (vku-booking-cache-v15) with self.skipWaiting() and clients.claim() for instantaneous activation.
- Integrated real-time matchMedia('(display-mode: standalone)') and appinstalled listeners to dynamically update the header button from 📱 Cài Đặt App to 🚀 Vào App and 🟢 Đang Trong App`.

Đà Nẵng, Ngày 04 tháng 10 năm 2026

Student Confirmation:

Nguyễn Thị Thương (Student ID: 23IT.B219 — Homeroom Class: 23SE4 — Course Class: Phát triển ứng dụng di động đa nền tảng (4))